FAMU-FSU College of Engineering

RFID Lab Access Control

Group 309

Introduction: Team 309

Members:

Patrick Laciuga	FSU, CPE	Project Manager
Zachary Gearhart-Robboy	FSU, EE	Hardware Lead
Calvin Reinert	FSU, CPE	Software Lead
Taylor Gooden	FAMU, CPE	Database Engineer
Michael Carter	FSU, CPE	Test Engineer
Jada Truong	FSU, CPE	Communications Lead

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Project Introduction

The Problem:

- Current student laboratories lack automated supervision, creating safety risks and making it difficult to track equipment usage.
- The FAMU-FSU College of Engineering requires a self-supervising system to act as a gatekeeper for potentially dangerous or expensive machinery.
- The system should not only track but ideally encourage responsible use of equipment

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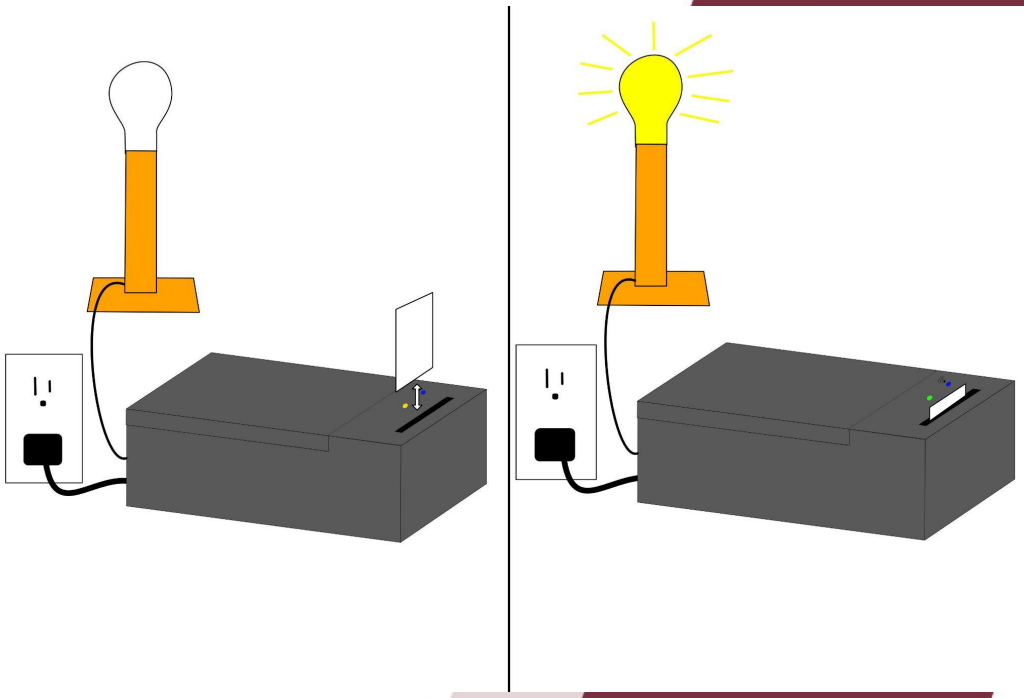
Project Introduction

Solution:

- An RFID-controlled power delivery system managed by a central database.

Key Objectives:

- Identify & Authenticate: Instantly verify authorized student IDs via RFID.
- Control Power: Safely deliver 120V/20A power to connected equipment contingent on authorization.
- Resist Tampering: Ensure the system cannot be easily bypassed to plug directly into the wall.
- Maintain Flow: Provide a simple, intuitive user interface that doesn't impede authorized students from getting their work done.



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Progress

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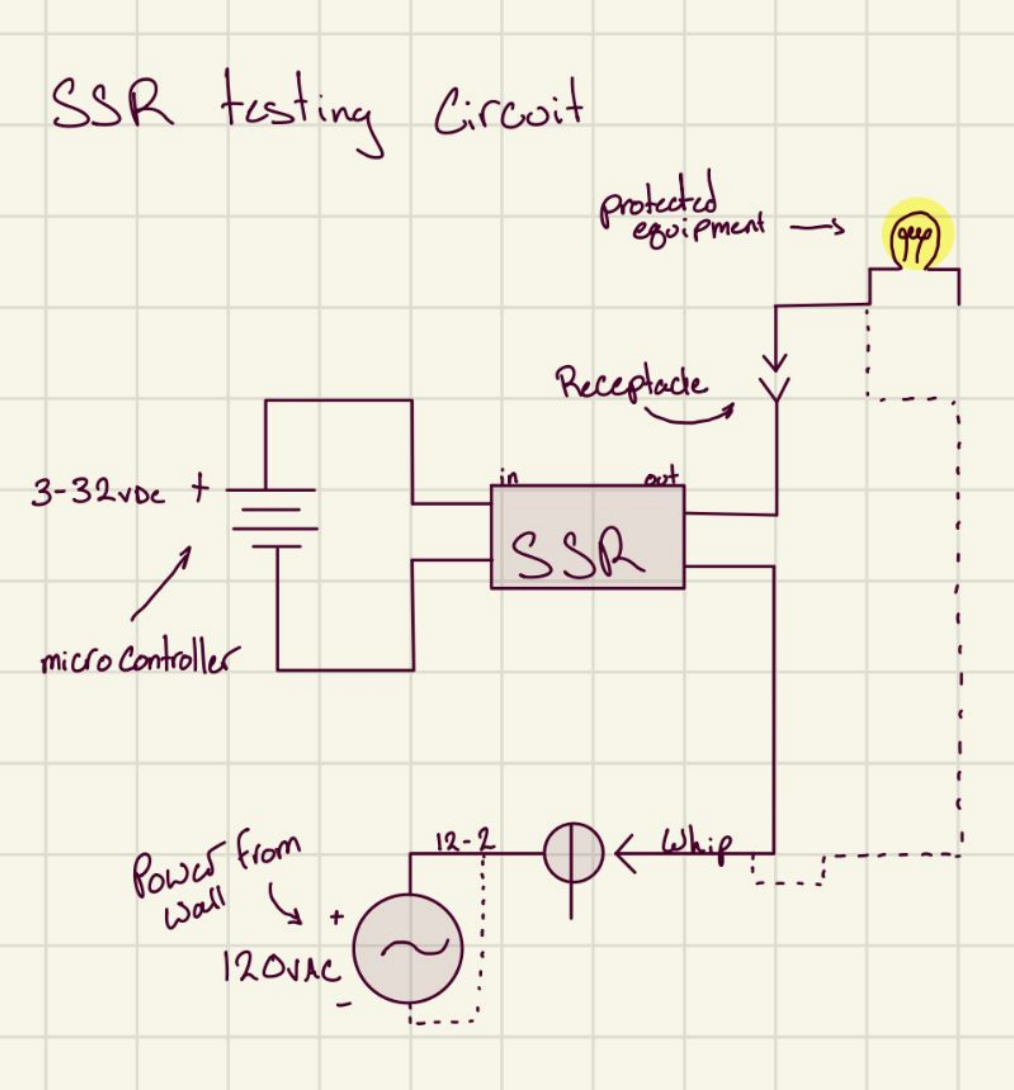
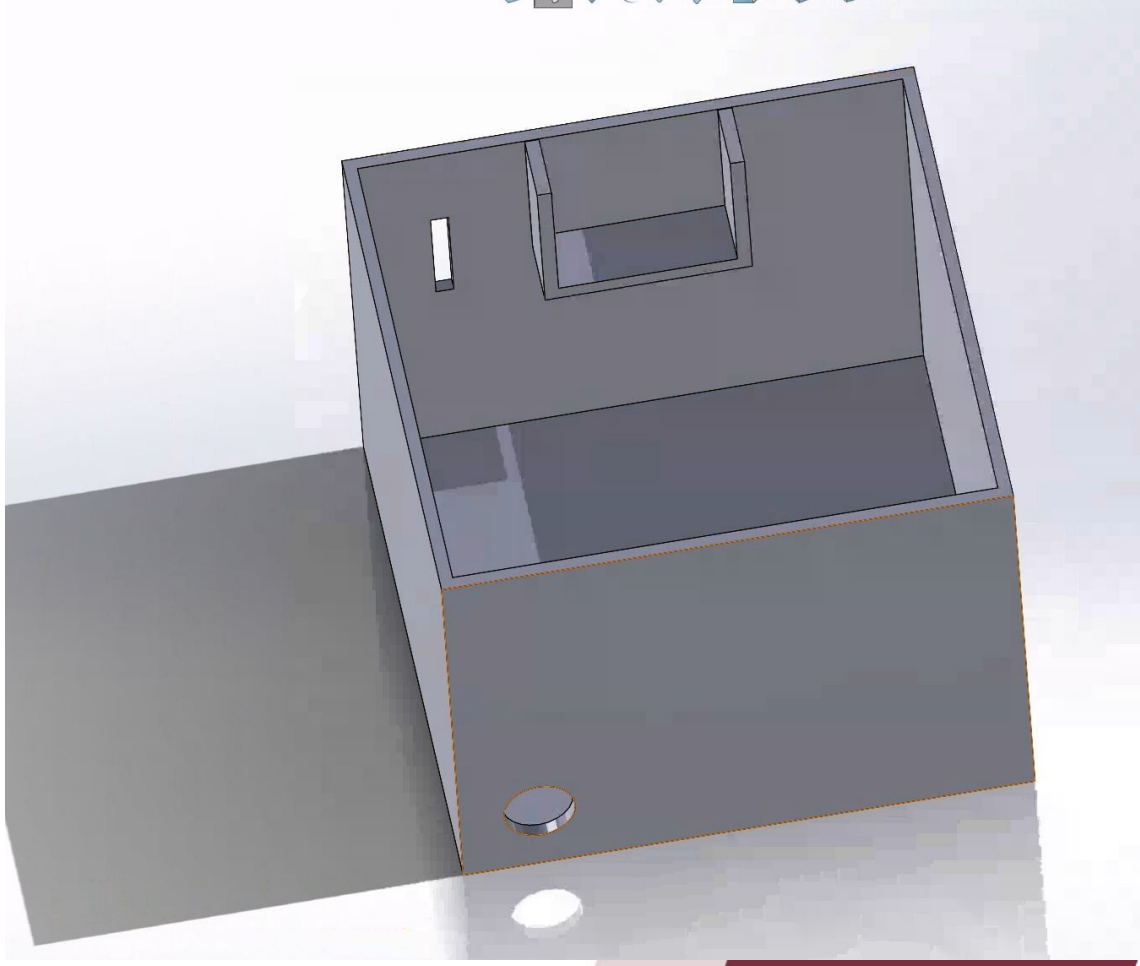
SSR Progress

- SSR operates within its parameters.
- Now, we need to connect the SSR to the microcontroller and test on a circuit with a high power draw.
- Need to figure out heat dissipation.



CAD design for box

- Small opening for LED strip.
- Shelf to hold the RFID reader.
  - Card will be inserted from the top
- 1 inch hole at the base for power cords.
- All components housed inside the box
- Currently working on printing a prototype.



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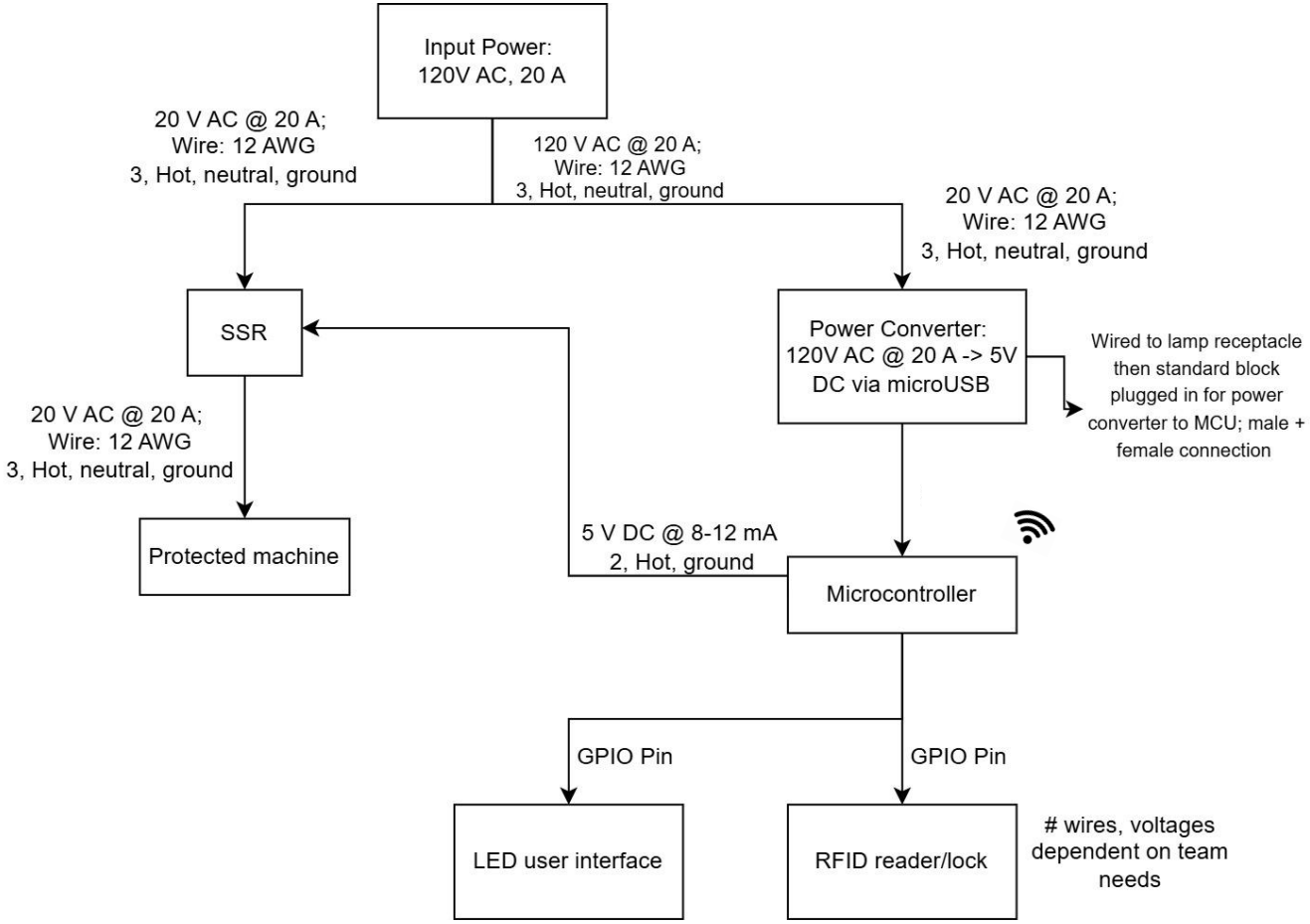
Zachary

General Circuit Review



Wires: 120V (>20A) -> 12awg
5V (>1A) -> 28 awg
Supports: Clamp on type
Power in: 20A Pigtail plug

Zachary

Digital Communication Flow

Information Source
Passive RFID/NFC tag UID

→

Input Transducer
PN532 performs 13.56 MHz RF modulation

→

Transmitter
PN532 RF field generator

↓

Channel
PN532 EM field

←

Receiver
PN532 front-end Demodulation occurs

←

Baseband Channel
PN532 Output is now input for Arduino via SPI

↓

Output Transducer
Arduino decoding for GPIO output actuation

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Jada

Code Algorithm Progress

Done:

- Arduino WiFi connectivity
- RFID card reading capability
- LED display for statuses

Next:

- Database connectivity
- SQL command communication & logging
- Power distribution

Calvin

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Code Examples:

Wifi Connection

```
while(wifiStatus != WL_CONNECTED){
  Serial.print("Connecting to WiFi...");
  wifiStatus = WiFi.begin(ssid, pass);
  delay(5000);
}
```

UID check

```
for (uint8_t i = 0; i < uidLength; i++) {
  Serial.print(" 0x"); Serial.print(uid[i], HEX);

  if(uid[i] == card1[i]){
    authenticated++;
  }
  else{
    authenticated = 0;
  }
}
```

This section will be replaced by a SQL query to the database

RFID card scan

```
// Look for an NFC card
success = nfc.readPassiveTargetID(PN532_MIFARE_ISO14443A, uid, &uidLength);
```

Wifi Details:

```
char ssid[] = SECRET_SSID; // ssid for connecting as client
char pass[] = SECRET_PASS; // pass for ssid
int APip[4] = {10,0,0,1}; // host ip when in AP mode
byte mac[6]; // mac address of arduino = 78:21:84:7A:2A:B4
WiFiServer server(80); // server port number
```

The ssid and password are stored in a separate “secrets.h” file for security

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Michael

Prototype Images

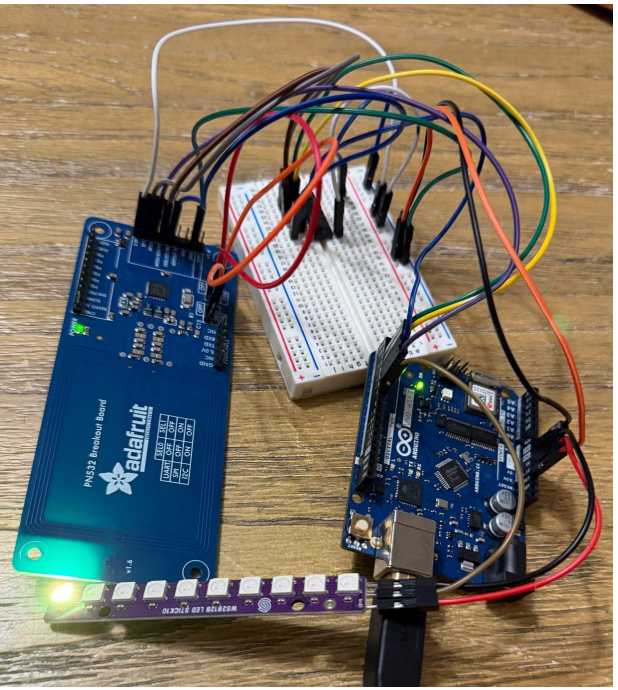


Image 1:

Wifi Status: Connecting

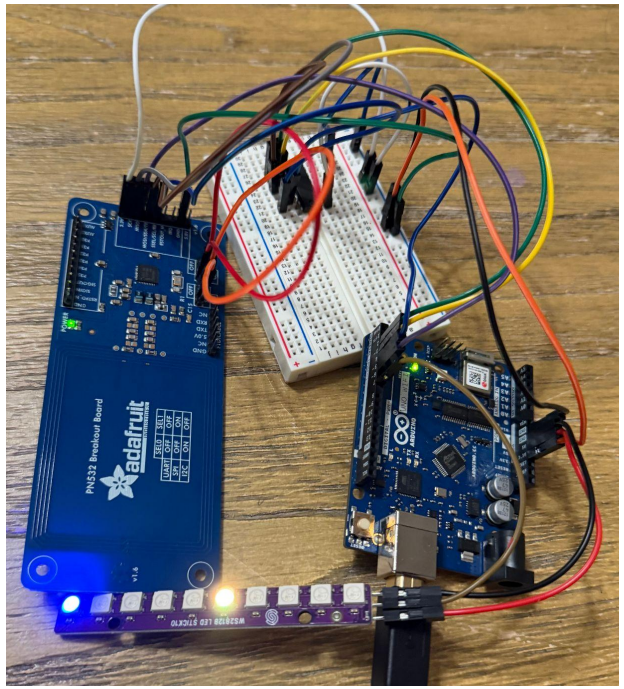


Image 2:

Wifi Status: Connected

Awaiting RFID card

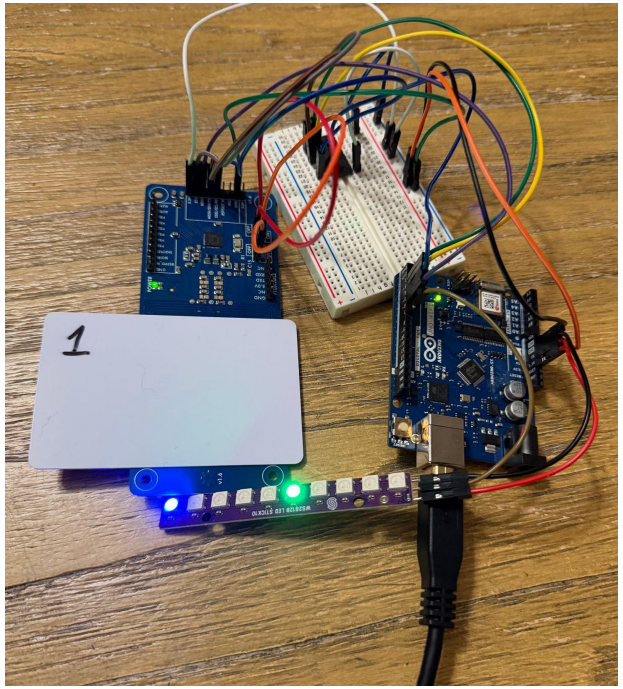


Image 3:

Valid RFID card scan

Remain GREEN until card is removed

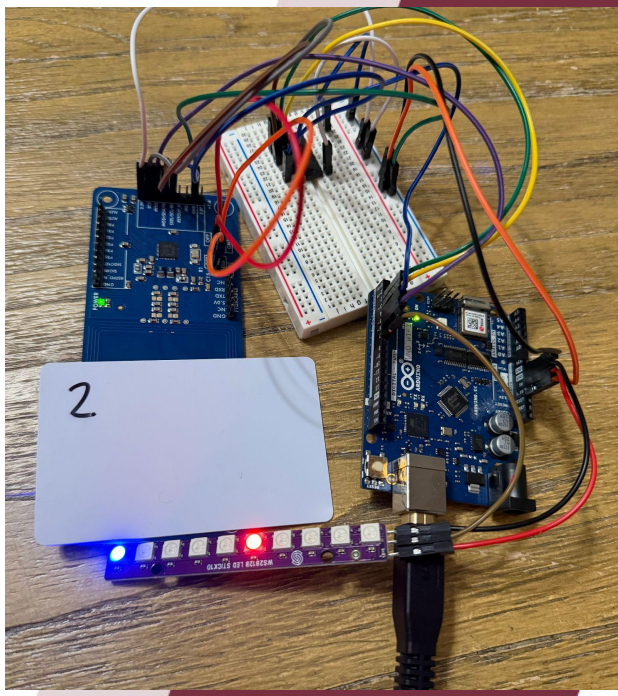


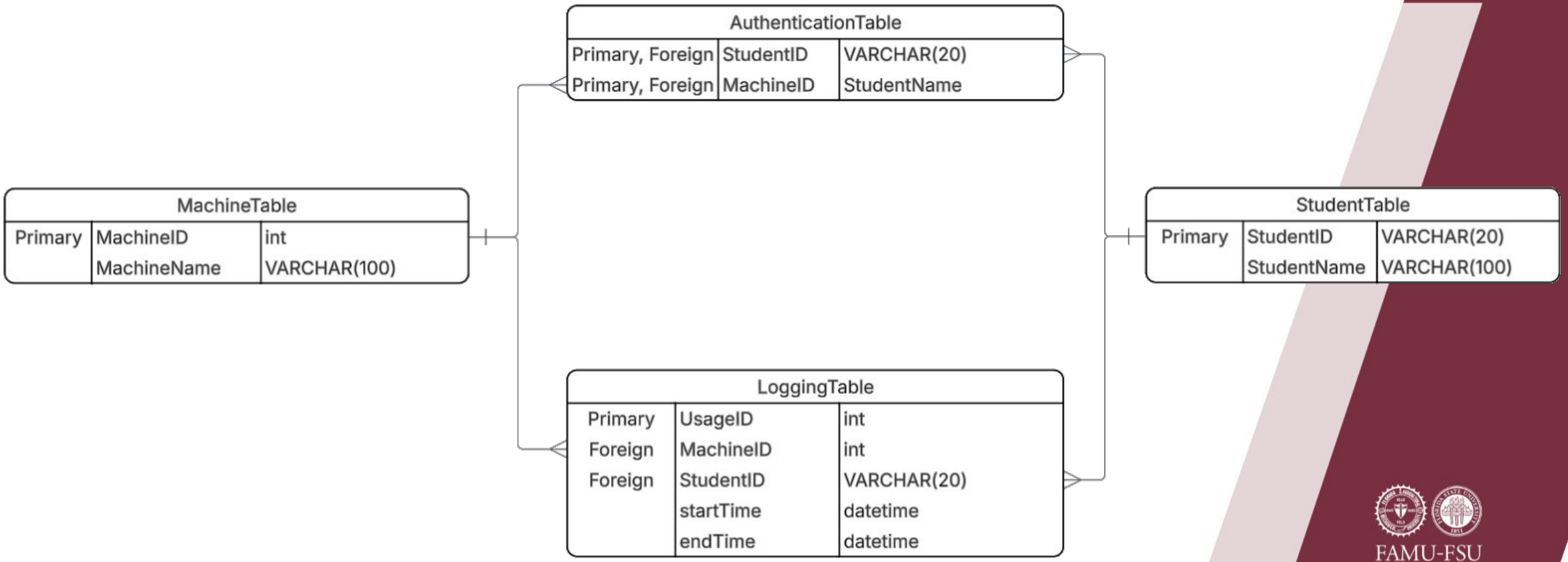
Image 4:

Invalid RFID card scan

Try again in 3 seconds

Michael

Database Schematic



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Website Progress

Current considered options

Option 1: Local Hosting

Web server running on a local machine, such as a raspberry pi.

Pi would require a static IP address for arduinos to connect.

Dependent on FSU ITS maintaining this IP

Would allow for an RFID reader to be directly connected for registering new student cards.

Issues with this machine would cause an outage across entire system.

Option 2: Cloud Hosting

Web server running through Microsoft Azure.

Arduinos would connect to a dynamic hostname.

Does not require FSU ITS to maintain IP

Would still require a dedicated machine and RFID reader to register new student cards.

More reliable - built upon Azure's infrastructure.

Web server is mostly transferable between options. Option 1 will be developed first as a prototype, and later transferred to option 2.

Next Steps

- Print and begin testing with 3D printed parts
- Connect with FSU IT to discuss networking
- Begin testing with a local database for arduino connectivity

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Jada